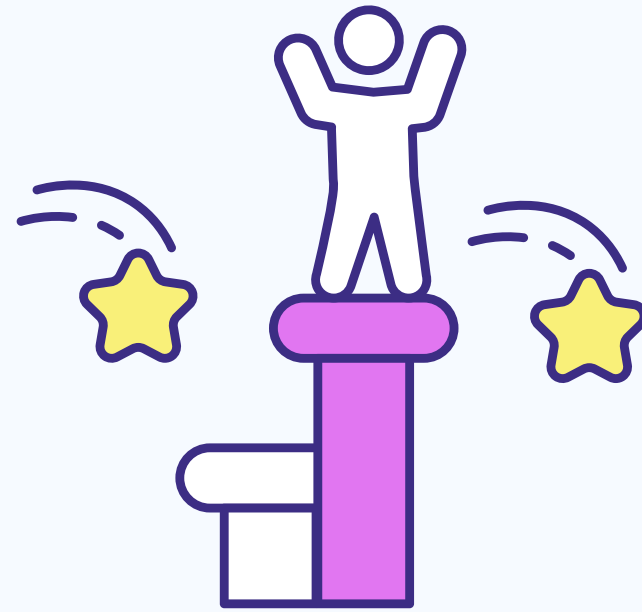




DEVOPS CORE

Intermediate Level

1 - DevOps Pre-Requisites

2 - DevOps Fundamentals

3 - DevOps Core

4 - DevOps Advanced



3 - DevOps Core - Part 1

Week 8

Weekly Progress Tracker

START Jenkins - 08.01: Intro to Build Automation

END Jenkins - 08.07: Freestyle to Pipeline Job

Key Topics for the Week

- **Jenkins in Docker:** Deploy Jenkins on a DigitalOcean droplet & run it in a Docker container
- **Jenkins UI Tutorial:** Navigating and using the Jenkins user interface
- **Gradle & Maven in Jenkins:** See how Gradle and Maven are used in Jenkins for building and managing Java projects
- **Jenkins Freestyle Jobs:** Learn how to create a freestyle job, connect it to a Git project, and execute shell commands within the job
- **Docker Integration:** Learn how to install Docker within Jenkins and how to run Docker commands in Jenkins
- **Jenkins Pipeline Jobs:** Understand the concept of Pipeline jobs as well as their benefits and use cases

Checklist for This Week

- ☐ Set up a DigitalOcean droplet, install Docker, and run a Jenkins container
- ☐ Configure Jenkins
- ☐ Install Gradle and Maven in Jenkins and verify setup
- ☐ Create a freestyle job connected to a Git project and execute shell commands
- ☐ Configure Jenkins volumes, install Docker, and integrate Docker build/login/push in Jenkins

Recommendations & Tips

- **Tip:** Ensure you pay close attention to indentation when creating and modifying your Jenkinsfile
- **Tip:** If you notice performance issues with your Jenkins server, and your allocated resources are sufficient - try updating any additional plugins to see if it improves performance
- **Remember:** If you are not completing the Bootcamp Exercises, you can destroy your DigitalOcean droplet that is hosting Nexus at the end of this stage to avoid any unnecessary charges

Week 9

Weekly Progress Tracker

START Jenkins - 08.08: Intro to Pipeline Job

END Jenkins - 08.15: Webhooks - Trigger Pipeline Jobs automatically

Key Topics for the Week

- **Creating a Jenkinsfile:** How a Jenkinsfile can be used to automate build, test, and deployment stages in a CI/CD pipeline
- **Advanced Jenkinsfile Configuration:** Using multiple stages in a Jenkinsfile, adding shell commands, credentials, and Groovy scripts to enhance pipeline functionality
- **Jenkins Credentials:** Explore the different types of credentials supported in Jenkins and when to use each (e.g., secret text, SSH keys, username/password)
- **Multibranch Pipeline:** How Jenkins can be configured to automatically build different branches from your Git repository
- **Groovy Scripts for Shared Libraries:** See how script.groovy files for Jenkins shared libraries can be used to centralize common pipeline code across multiple projects
- **GitLab Integration:** How to use the GitLab plugin to configure webhooks for automatic pipeline triggers when changes are pushed to the GitLab repository

Checklist for This Week

- ☐ Write a Jenkinsfile to automate build, test, and deployment processes
- ☐ Implement multiple stages in the Jenkinsfile, including shell commands and Groovy scripting
- ☐ Create a Jenkins pipeline with defined credentials
- ☐ Set up a multibranch pipeline to handle multiple branches of your Git project
- ☐ Write and test a script.groovy for a shared library in Jenkins
- ☐ Install and configure the GitLab plugin in Jenkins, and add a webhook to trigger pipelines automatically

Recommendations & Tips

- **Tip:** Make sure the name for your maven installation in the Jenkinsfile matches the name you have set in the Jenkins dashboard to avoid any errors!

Week 10

Weekly Progress Tracker

START Jenkins - 08.16: Dynamically Increment Application

version in Jenkins Pipeline - Part 1

END AWS - 09.05: VPC - Manage Private Network on AWS

Key Topics for the Week

- **Pom.xml file:** How to use a pom.xml file to automate version increments during builds
- **Jenkinsfile Version Management:** How to use a Jenkins pipeline to increment the version, commit, and push changes
- **AWS Introduction:** Understanding AWS' UI, key services such as IAM and VPC and the Free Tier
- **AWS Global Infrastructure:** AWS regions, availability zones

Checklist for This Week

- ☐ Configure pom.xml to increment the java-app project version automatically during Jenkins builds
- ☐ Update Jenkinsfile to handle version increments, commit, and push to Git
- ☐ Explore the Jenkins UI and understand key services in pipeline management
- ☐ Set up an IAM user, configure roles and groups

Recommendations & Tips

- **Reminder:** Keep an eye on your free-tier usage in AWS to prevent accidental charges



3 - DevOps Core - Part 1

Week 11

Weekly Progress Tracker

START AWS - 09.06: CIDR Blocks explained

END AWS - 09.14: Container Services on AWS Preview

Key Topics for the Week

- **CIDR Blocks:** Understand CIDR notation, subnetting, and how to use subnet calculators for network planning
- **EC2 Instance management:** Instances, key pairs & security groups
- **Jenkins & Docker Integration with EC2:** How to increment your app's version, build and push an application image and deploy to EC2 using a Jenkins pipeline
- **AWS CLI:** How AWS CLI commands can be used to manage your infrastructure
- **Introduction to IaC and Microservices:** Understanding Infrastructure as Code (IaC) principles and the concept of microservices, and Amazon EKS for container orchestration

Checklist for This Week

- ☐ Use a subnet calculator to understand CIDR and subnetting
- ☐ Create an EC2 instance, configure SSH, install Docker, pull an image, and run the app
- ☐ Add the EC2 SSH key to Jenkins and update the Jenkinsfile with Docker and EC2 commands
- ☐ Install Docker Compose on EC2, create a docker-compose.yml file, and copy it using Jenkins
- ☐ Install and configure AWS CLI, create a security group, and launch an EC2 instance using CLI commands

Recommendations & Tips

- **Reminder:** Ensure correct permissions are set on the EC2 SSH key to avoid any errors
- **Tip:** Check this module's Git repository for a list of useful AWS CLI commands



3 - DevOps Core - Part 2

Week 12

Weekly Progress Tracker

START K8s - 10.01: Intro to Kubernetes

END K8s - 10.09: Services - Connecting to Application..

Key Topics for the Week

- **Introduction to Containers and Orchestration:** Understand the need for containers and explore features of container orchestration tools like Kubernetes.
- **Kubernetes Components:** Learn about nodes, pods, services (internal and external), volumes, deployments, statefulsets, replicas, and daemonsets
- **Kubernetes Architecture:** Understand key Kubernetes elements: worker nodes, container runtime, kubelet and control plane processes
- **Kubectl & Minikube:** Introduction to kubectl and Minikube
- **Using Kubectl:** Essential kubectl commands like *get nodes*, *create/edit deployments*, *get logs*, *describe resources*
- **YAML Configuration in Kubernetes:** Service and deployment structure, specifications, statuses, indentation rules, and port configuration
- **Kubernetes Namespaces and Services:** Kubernetes namespaces and service types like ClusterIP, NodePort, and LoadBalancer

Checklist for This Week

- ☐ Install Minikube and Docker Desktop, and run basic kubectl commands for managing clusters
- ☐ Write and deploy a Kubernetes YAML file, ensuring correct indentation and syntax
- ☐ Deploy MongoDB with secrets, services, and a Mongo Express deployment using a LoadBalancer

Recommendations & Tips

- **Reminder:** Pay close attention to YAML indentation when writing Kubernetes configurations as errors can lead to failed deployments.

Week 13

Weekly Progress Tracker

START K8s - 10.10: Ingress - Connecting to Applications outs..

END K8s - 10.18: Kubernetes Operators for Managing Com..

Key Topics for the Week

- **Ingress & Routing Rules:** Learn about Ingress, routing rules, and Ingress controllers.
- **Minikube:** Explore Minikube Dashboard and Minikube tunnel, and understand TLS certificates for secure communication.
- **Persistent Storage in Kubernetes:** Explanation of Persistent Volumes (PVs), Persistent Volume Claims (PVCs), ConfigMaps, Secrets, volume types, and storage classes.
- **Stateful Sets and Scaling:** Explore StatefulSets in Kubernetes, focusing on scaling database applications and managing stateful workloads.
- **Managed vs Unmanaged Clusters:** Understand the differences between managed and unmanaged Kubernetes clusters, data persistence, and load balancing considerations
- **Helm and Package Management:** Learn about Helm, charts, the Artifact Hub, templating and values, and how to manage releases using Helm
- **Kubernetes Operator:** What it is, why and when it is used

Checklist for This Week

- ☐ Configure Minikube dashboard and tunnel
- ☐ Create a Mosquitto deployment using volumes and a secret file for configuration.
- ☐ Compare managed and unmanaged Kubernetes clusters, focusing on load balancing and data persistence
- ☐ Install a Bitnami Helm chart in an LKE cluster, including MongoDB, Nginx Ingress Controller, and Mongo Express
- ☐ Deploy images from a private docker repository - create a secret, authenticate with AWS ECR, and deploy from ECR to your cluster

Recommendations & Tips

- **Reminder:** If you are having any issues with DigitalOcean or Linode accounts being blocked - please contact our support team immediately. You can use AWS for these tasks as a backup if it is absolutely necessary

Week 14

Weekly Progress Tracker

START K8s - 10.19: Secure your cluster - Authorization with ..

END K8s - 10.24: Demo project: Deploy Microservices with ..

Key Topics for the Week

- **Kubernetes RBAC and Role Configuration:** Roles, RoleBindings, and RBAC (Role-Based Access Control) in Kubernetes. Learn about users, groups, external authentication, API server communication, and application authentication
- **Microservices:** Introduction to microservices, their benefits, and how services communicate with each other within a distributed system
- **Linode Kubernetes Engine:** How to deploy microservices on Linode Kubernetes Engine (LKE).
- **Kubernetes Best Practices:** Learn about version management, health checks, liveness probes, resource limits, and load balancers for microservices, as well as vulnerability scanning
- **Helm Charts and Deployments:** How to use YAML configuration files to standardize deployments, a Helmfile to manage multiple charts, and where to host and manage Helm charts for future use.

Checklist for This Week

- ☐ Create a YAML file for microservices, including variables and ports, and deploy them on LKE
- ☐ Set up health checks, liveness probes, and resource limits for microservices, and ensure proper load balancing
- ☐ Create a Helm chart with appropriate values and configuration, and deploy it to LKE
- ☐ Create a Helmfile to manage multiple deployments and understand where to host Helm charts for reuse

Recommendations & Tips

- **Tip:** When configuring RBAC, start with least-privileged access to enhance security within your cluster.



3 - DevOps Core - Part 3

Week 15

Weekly Progress Tracker

START AWS - 11.01: Container Services on AWS

END EKS - 11.04: Create Fargate Profile for EKS Cluster

Key Topics for the Week

- **Containers on AWS:** Overview of container services on AWS—ECS, Fargate, and EKS. Understand the pros and cons of each service and compare their use cases.
- **EKS Setup:** Learn how to use IAM roles and VPCs worker nodes using CloudFormation to create an EKS cluster. Understand how EC2 roles can be used to set up a node group to manage compute resources.
- **Autoscaling in EKS:** Understand autoscaling in an EKS cluster, how to review autoscaling logs, and ensure resource efficiency, and how a load balancer can be used to manage traffic distribution.
- **Fargate:** See how to use Fargate for running serverless containers

Checklist for This Week

- ☐ Compare ECS, Fargate, and EKS, understanding their pros, cons, and use cases
- ☐ Create an EKS IAM role, VPC, EC2 role, and node group using CloudFormation
- ☐ Configure autoscaling for EKS, monitor logs, and set up a load balancer for traffic management
- ☐ Create a Fargate role and profile, and run serverless containers using Fargate
- ☐ Safely terminate the EKS cluster, node groups, and Fargate infrastructure to avoid ongoing costs

Recommendations & Tips

- **Tip:** Ensure you are creating the EKS cluster role with the correct AWS account to avoid any permissions issues

Week 16

Weekly Progress Tracker

START EKS - 11.05: Create EKS cluster with eksctl command line tool

END EKS - 11.10: Complete CI/CD Pipeline with EKS and ECR

Key Topics for the Week

- **EKSCTL Setup and Infrastructure:** Learn how the eksctl command line tool can be used to provision infrastructure in AWS
- **Deployment to EKS and LKE from Jenkins:** How to deploy to EKS and LKE using Jenkins pipeline
- **Jenkins Credential Management Best Practices:** Explore best practices for managing credentials securely in Jenkins and Kubernetes
- **Complete CI/CD Pipeline:** How to configure your environment to run a complete CI/CD pipeline using EKS, DockerHub and ECR

Checklist for This Week

- ☐ Install eksctl and recreate the EKS infrastructure from previous lectures using the command line
- ☐ Configure AWS IAM Authenticator, kubectl, and kubeconfig in Jenkins, and update the Jenkinsfile for EKS deployment
- ☐ Create an LKE cluster, configure Jenkins for LKE authentication, and update the Jenkinsfile
- ☐ Update the deployment YAML and Jenkinsfile to deploy from DockerHub using Jenkins
- ☐ Create an ECR repository, set up credentials in Jenkins, and deploy the application from ECR

Recommendations & Tips

- **Tip:** You can delete the infrastructure created during this weeks lectures to ensure no excess AWS charges are incurred